

STEMLens Lab

AI-generated STEM micro-labs that turn confusion into prediction, simulation, explanation, and feedback.

Problem

Many students struggle in STEM because formulas arrive before the underlying behavior is visible. They may memorize projectile motion, Ohm's law, or $y = mx + b$ without a testable mental model of what each variable does.

Solution

STEMLens Lab asks what feels confusing, creates a short learning mission, opens an interactive simulation, and gives concise feedback after the student explains what changed. It is designed as a misconception repair loop rather than a generic chatbot.

Prototype Features

Personalized mission	Hugging Face/DeepSeek generation when configured, deterministic fallback for reliable demos.
Interactive labs	Projectile motion sandbox, Ohm's law circuit meter, and linear equation graph explorer.
Feedback loop	Students predict, manipulate sliders, explain observations, and receive targeted feedback.
Accessible design	Visual-first custom interface for learners who need concepts to move before formulas click.

Technical Implementation

FastAPI backend, custom HTML/CSS/JavaScript frontend, canvas simulations, Hugging Face inference support, and fallback lesson templates. The prototype remains usable even if model credentials or inference capacity are unavailable during judging.

Impact

The project supports self-study, tutoring, and classroom review by helping students build visual intuition. It is especially useful for learners who do not benefit from static explanations alone.